គណនាអាំងតេក្រាល

$$I = \int \frac{\cos^3 x}{\sin^3 x} dx$$

$$k = \int \frac{1}{(\sin x + \cos x)^2} dx$$

Ron Chanratt

ដំណោះស្រាយ

$$I = \int \frac{\cos^3 x}{\sin^3 x} dx$$

$$= \int \frac{\cos x \,.\, (1 - \sin^2 x)}{\sin^3 x} dx$$

តាង: $u = sinx$

$\rightarrow du = \cos x \, dx \;\; \rightarrow dx = \frac{du}{\cos x}$

$$= \int \frac{\cos x \,.\, (1 - u^2)}{u^3} \cdot \frac{du}{\cos x}$$

$$= \int \frac{1}{u^3} du - \int \frac{1}{u} du = -\frac{1}{2u^2} - ln|u|$$

$$= -\frac{1}{2\sin^2 x} - ln|\sin x| + c$$

ដូចនេះ

$$\boxed{I = -\frac{1}{2\sin^2 x} - ln|\sin x| + c}$$

$$k = \int \frac{1}{(\sin x + \cos x)^2} dx$$

ចំណាំ:

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$\sin 2x = 2 \sin x \cos x$$

$$\sin^2 x + \cos^2 x = 1$$

$$= \int \left(\frac{1}{\sin^2 x + 2 \sin x \cos x + \cos^2 x} \right) dx$$

$$= \int \frac{1}{1 + \sin 2x} dx$$

តាង: $u = 2x$

$\rightarrow du = 2dx \quad \rightarrow dx = \frac{du}{2}$

$$= \frac{1}{2}\int \frac{1}{1+\sin u}\,du$$

តាង: $t = \tan\frac{u}{2}$

$\rightarrow dt = \frac{1}{2}(1+t^2)du \quad \rightarrow du = \frac{2}{1+t^2}dt$

$$= \frac{1}{2}\cdot 2\int \frac{1}{1+\frac{2t}{1+t^2}}\cdot\frac{dt}{1+t^2}$$

$$= \int \frac{1}{(t+1)^2}\,dt$$

តាង: $t+1 = s \quad \rightarrow dt = ds$

$$= \int \frac{1}{s^2} ds = -\frac{1}{s}$$

$$= -\frac{1}{\tan\left(\frac{u}{2}\right) + 1}$$

$$= -\frac{1}{\tan(x) + 1} + c$$

ដូចនេះ

$$\boxed{k = -\frac{1}{\tan(x) + 1} + c}$$